

Incremental Typesetting

Why a from-scratch engine can be fast

J. Whitfield

FlashTeX corpus

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A slide deck is a sequence of frames. Each frame is a page of paper that is 128 mm wide and 96 mm tall, typeset in a sans-serif face with no paragraph indentation and a small skip between paragraphs.

This deck has six content frames after the title page. The text on this page is deliberately long enough to wrap onto several lines so that the line breaks inside a frame can be measured against the reference.

The problem

- Every keystroke re-runs the whole document.
- A page that did not change is typeset again anyway.
- Fonts are loaded from scratch on every run.
- The result is a preview that lags the editor by seconds.

What we keep

- ① The token stream of every paragraph, keyed by its source span.
- ② The shaped glyph runs of every unchanged paragraph.
- ③ The page breaks up to the first changed page.

Only the paragraph under the cursor is reshaped; the rest is replayed.

The compiler produces a paragraph list. The renderer breaks lines with the Knuth–Plass algorithm and builds pages. The exporter writes the PDF.

- compiler: tokens $\rightarrow$ paragraphs
- render pipeline: paragraphs $\rightarrow$ pages
- pdf: pages $\rightarrow$ bytes

We compare word positions against pdfLaTeX at 144 dpi. A page counts as matching when every aligned word is within one point of its reference position and the differing pixels are below one percent.

- 1 Render both documents.
- 2 Align the words by text.
- 3 Report the largest deltas first.

Summary

Caching the paragraph layer gives most of the win; caching pages gives the rest. The oracle stays outside the product.